

Hi

In this segment, we will have a look at value iteration algorithm.

We will import the libraries and make the 4x4 gridworld environment.

Recall the bellman optimality equation for value function.

If we start with an arbitrary value function and iteratively apply this equation, we eventually get the optimal value function.

From that value function, we can get the optimal policy by acting greedily w.r.t. It.

From another perspective, recall that in policy iteration, we evaluate the policy till convergence.

If we only do the first iteration of evaluation and then directly proceed with the improvement step, we get the value iteration algorithm.

As for the code, `one_step_lookahead` is same as before.

We start with the value function which is zero everywhere. Then, for each state, we assign the maximum return according to the current value function. We stop when the change becomes smaller than a pre-defined threshold.

Then, we construct the optimal policy.

Now, let's run the algorithm. Next, we check that the value function is close to the expected one.

Then, we plot both the value functions.

Now let's see the optimal policy in action.

It manages to reach the terminal state at the top.

To summarize, we constructed the value iteration algorithm from the bellman optimality equation for value function  $V$ . Later, we will see which algorithm emerges from the optimality equation for  $Q$ . Thank you.